

VR Everywhere - Virtual Travel Experience

Project Proposal for UCS503P
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July – December 2026

1 Higher-order goal

...immersive travel experience

This project aims to make travel exploration more accessible by allowing users to virtually experience tourist destinations without physically travelling to them. While the broader goal is to explore the potential of virtual reality in tourism and education, the immediate engineering objective is to develop a simple, deployable VR-based travel application that allows users to explore selected destinations through immersive environments.

The project will focus on providing an easy-to-use interface, destination information, interactive exploration, and basic VR navigation rather than attempting to create a complete commercial tourism platform.

2 Time-to-value

...primary heuristic

- We will deliver meaningful increments quickly by first implementing the core workflow: selecting a destination → viewing its information → starting a VR tour → exploring the environment.
- We will prioritize a small number of high-quality destinations over attempting to build a large virtual world.
- Development will follow short weekly iterations, with each iteration producing a demonstrable feature or improvement.
- Testing will be performed continuously during development rather than being postponed until the final stage.
- Success will initially be defined by whether a user can successfully discover a destination and complete a basic VR tour, followed by improvements based on user feedback.

3 Problem Statement

Travelling to tourist destinations is not always possible for everyone due to cost, time constraints, physical limitations, geographical distance, or other circumstances.

- **Accessibility:** People who cannot physically travel may have limited opportunities to experience tourist destinations.

- **Cost:** International and long-distance travel can be expensive.
- **Time constraints:** Students and working individuals may not have sufficient time to visit multiple destinations.
- **Limited pre-travel experience:** Traditional travel websites provide photographs and videos, but they do not provide an immersive experience of the location.
- **Information fragmentation:** Information about a destination, its history, attractions, and visual experience is often spread across multiple platforms.

Virtual Reality provides an opportunity to address these limitations by allowing users to experience selected tourist destinations in an immersive environment.

4 Proposed Solution

...Software Proposition

4.1 Overview

Build a VR-based “Virtual Travel Experience” system with the following components:

- **Destination explorer:** Users can browse and search for available tourist destinations.
- **Destination information:** Users can view basic information such as location, description, historical facts, and important attractions.
- **VR tour:** Users can enter a virtual environment and explore a selected destination.
- **Interactive hotspots:** Important locations within the VR environment can be selected to display additional information.
- **Audio guide:** Basic audio narration can be provided for selected attractions.
- **Favorites:** Users can save destinations for later exploration.
- **User profile:** Users can view their saved and previously explored destinations.

The initial implementation will contain a limited number of destinations to keep the project achievable within the three-month development period.

4.2 Core workflow

1. User opens the application and logs in or continues as a user.
2. User views available destinations on the Home/Explore screen.
3. User selects a destination.
4. The system displays destination information and provides a Start VR Tour option.
5. User enters the VR environment and explores the destination.
6. User can interact with selected hotspots to view information or listen to an audio guide.
7. User exits the tour and can save the destination to Favorites.

This keeps the central workflow simple and demonstrable.

4.3 Operational constraints for an educational project

- The application will initially support a small number of destinations, approximately 3–5, rather than attempting to model the entire world.
- VR environments will use available 360° images/assets or lightweight 3D environments instead of developing highly detailed environments from scratch.
- The application should remain usable on supported VR hardware and provide a basic non-VR/desktop mode where feasible for demonstration.
- Advanced features such as multiplayer VR, live tour guides, complex AI systems, and real-time environmental simulation are outside the initial scope.
- The system will use commonly available development tools and cloud hosting services.

5 Solution Approach

...Engineering focus

This is an engineering course project, so the primary focus will be on system design, VR interaction, usability, integration, testing, and deployment, rather than developing new VR algorithms or conducting research into virtual reality.

5.1 VR environment and interaction

...practical implementation

- Basic VR navigation.
- 360° environment viewing.
- Interactive information hotspots.
- Audio narration.
- Simple VR menu.
- Destination exit/return functionality.

The project will prioritize usability and stable interaction over highly complex graphical effects.

5.2 Application architecture

...web-connected VR solution

A typical architecture will consist of:

- **VR Client:** Unity-based application responsible for the VR environment, navigation, interaction, and user interface.
- **Backend API:** Responsible for authentication, destination information, user preferences, favorites, and tour history.
- **Database:** Stores users, destinations, hotspot information, favorites, and tour history.
- **Cloud/Deployment Layer:** Hosts the backend and database and provides access to the application services.

The architecture separates the VR application from backend services so that destination information can be updated without modifying the complete application.

5.3 Development and deployment

...fast delivery enabler

- Use Git and GitHub for version control.
- Automatically build and test the backend when changes are pushed, where CI/CD is introduced.
- Run backend unit tests.
- Maintain a consistent deployment process.
- Package the VR application as a suitable build for the selected demonstration platform.

6 Evaluation Criterion

...measurable and attributable

6.1 Primary evaluation metric

Tour Completion Rate: Percentage of test users who can successfully select a destination, enter the VR tour, interact with a hotspot, and exit the tour without assistance.

- **Target:** At least 80% of test users should successfully complete the workflow during user testing.

6.2 Secondary evaluation metrics

- **Usability:** Average user rating of the application interface on a 1–5 scale.
- **Navigation success:** Percentage of users who can successfully navigate or interact within the VR environment.
- **Performance:** Maintain a stable and comfortable VR experience on the selected hardware.
- **Feature usage:** Number of users who interact with information hotspots or audio guides.
- **Reliability:** Successful loading of destinations and VR environments during testing.

6.3 Pilot validation plan

...high-level

- Recruit approximately 10–20 student users for testing.
- Provide users with a simple task such as: “Find the Taj Mahal, start the VR tour, explore the location, open an information hotspot, and exit the tour.”
- Record whether users complete the task successfully.
- Collect feedback using a short questionnaire.
- Identify usability problems and improve the interface before final deployment.

7 Scalability

...with foresight

7.1 Scalable theoretically

- Store destination information in the database.
- Use separate VR scenes/environments for destinations.
- Separate the frontend/VR client and backend services.

- Use indexed database queries for destination searches.
- Keep backend APIs stateless where practical.

7.2 Operationally deployable

- Cloud-hosted backend.
- Managed relational database.
- Version-controlled source code.
- Repeatable deployment process.

This allows the initial 3–5 destinations to eventually grow into a larger virtual tourism catalogue.

8 Engine availability heuristic

- **VR/Game Engine:** Unity
- **Programming Language:** C#
- **VR Framework:** OpenXR + XR Interaction Toolkit
- **Backend:** Spring Boot
- **Backend Language:** Java
- **Database:** PostgreSQL
- **Authentication:** Spring Security + JWT
- **API:** REST API
- **API Testing:** Postman
- **Version Control:** Git + GitHub
- **Deployment:** Render / Railway / similar cloud platform

These technologies allow the project to focus on VR interaction, application workflow, testing, and deployment rather than developing infrastructure from scratch.

9 Project scope and deliverables

...iteration-friendly

9.1 Initial deliverable

...first iteration

- Basic application interface.
- Home/Explore screen.
- Destination listing.
- Destination details page.
- One working VR destination.
- Basic VR navigation.

- At least one interactive information hotspot.
- Basic backend API.
- Database containing destination information.
- Git repository and basic testing.

9.2 Subsequent deliverables

- Additional VR destinations.
- Search functionality.
- Audio guide.
- User profile.
- Tour history.
- Improved VR interface.
- Backend integration.
- System and user testing.
- Performance optimization.

10 Risks and mitigations

Risk	Mitigation
VR hardware availability	Develop and test on the available VR device and maintain a desktop mode where feasible for demonstration.
Complexity of VR development	Limit the initial project to 3–5 destinations and basic interactions.
Performance issues	Use optimized 3D assets, appropriate texture sizes, and lightweight environments.
Limited development time	Follow weekly iterations and prioritize the core VR tour workflow.
Asset availability	Use appropriately licensed/free assets and 360° environments rather than attempting to create every asset from scratch.
User discomfort	Keep movement simple, provide comfortable navigation options, and test the application with users.
Deployment difficulties	Deploy backend services early rather than waiting until the final week.

11 Summary

This project proposes a deployable VR Travel Experience platform that allows users to explore selected tourist destinations through immersive virtual environments. The system will combine a simple destination discovery interface with VR tours, interactive hotspots, basic destination information, and user favorites.

The project meets the course requirements by emphasizing rapid development, structured software engineering, measurable user evaluation, testing, and deployment. The initial scope is intentionally limited to a small number of destinations and essential VR interactions so that a functional and polished system can be completed within three months.

The architecture will also allow additional destinations and features to be introduced in future iterations without redesigning the complete system.

Technology Stack

Component	Technology
VR Application	Unity
Programming	C#
VR Framework	OpenXR + XR Interaction Toolkit
Backend	Spring Boot
Backend Language	Java
Database	PostgreSQL
Authentication	Spring Security + JWT
API	REST API
API Testing	Postman
Version Control	Git + GitHub
Deployment	Render / Railway / similar cloud platform

Core User Interface

- The main application navigation will contain five simple options: **Home** | **Explore** | **VR Tours** | **Favorites** | **Profile**
- Inside the VR tour, a lightweight floating menu will provide: **Map** | **Information** | **Audio** | **Settings** | **Exit Tour**